Q1

a) Define the following:

i. Web Browser:

A web browser is an application software used to access and view websites on the Internet.

Examples: Google Chrome, Mozilla Firefox, Microsoft Edge.

ii. Switches:

A switch is a networking device that connects multiple devices in a LAN and forwards data only to the device that needs it, using MAC addresses.

iii. Routers:

A router is a device that connects multiple networks together and directs data packets between them using IP addresses.

iv. IP Configuration:

IP Configuration refers to the process of assigning IP addresses, subnet masks, gateways, and DNS servers to a device in a network.

Command (Windows): ipconfig

Q2

a) Explain Ethernet Data Link Protocol. (6 Marks)

The Ethernet Data Link Layer is divided into two sub-layers:

1. Logical Link Control (LLC): Manages communication between devices and controls frame synchronization.

2. Media Access Control (MAC): Responsible for addressing and access control to the physical medium.

Ethernet Frame Fields:

Preamble - Synchronization bits

Destination MAC Address

Source MAC Address

Type/Length

Data

Frame Check Sequence (FCS)

b) Describe SOHO LAN. (3 Marks)

SOHO (Small Office Home Office) LAN is a small-scale local area network used in small businesses or home offices.

Connects few computers, printers, and routers.

Provides file sharing, Internet access, and data storage.

Inexpensive and easy to manage.

Q3

a) Analyze Network Layer Protocols with a Type Field. (4 Marks)

Network layer protocols include:

IP (Internet Protocol): Provides logical addressing and routing.

ICMP (Internet Control Message Protocol): Used for error reporting and diagnostics.

ARP (Address Resolution Protocol): Maps IP address to MAC address.

Type Field: Specifies which protocol is being used (e.g., 0x0800 for IPv4).

b) Discuss Hierarchical Network Design. (4 Marks)

Hierarchical design divides the network into three layers:

1. Core Layer: High-speed backbone connecting distribution layers.

2. Distribution Layer: Routing, filtering, and policy control.

3. Access Layer: Connects end devices (PCS, printers).

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Advantages: Scalable, y management,

Discuss IP Addressing & how it helps IP Routing.

IP Addressing: Assigns a unique address to every device on a network.

Example: 192.168.1.1

IP addresses are divided into Network ID and Host ID.

Helps routing by allowing routers to identify destination networks and forward packets correctly.